Employee Absenteeism Analysis and Prediction

IT3081 Statistical Modelling Consultancy Report

Human Resources and Manufacturing

Dataset: MFGEmployees4.csv | 8,336 employee records | 13 observed variables

Submission note: This report contains the reproducible statistical analysis, literature review, innovation proposal, and recommendation framework. Industry expert validation remains pending until a real consultation is completed and documented.

Employee Absenteeism Analysis and Prediction

Consultancy Report Draft

This document is a structured evidence report. Numeric findings must be taken from outputs/ after running python main.py. Do not replace placeholders with invented results.

1. Executive Summary

Business problem: Employee absence can reduce productivity and increase staffing and operational pressure.

Data: The supplied file contains 8,336 records and 13 variables. It has no missing values or duplicate employee identifiers. Quality concerns include 34 implausibly young ages and one service-age inconsistency.

Current evidence: AbsentHours has mean 61.284, median 56.006, standard deviation 49.038, and 15.83% zero values. Age has a strong observed Pearson correlation with AbsentHours (r = 0.8302), but this is observational and should be checked for data-generation artefacts and sensitivity to flagged ages.

Recommendation: Complete the generated inference and model tables, select a model using predictive performance, assumptions, stability, and management interpretability, and use outputs for supportive workforce planning rather than automated individual sanctions.

2. Introduction

2.1 Industry background

Discuss manufacturing/Human Resources workforce continuity, staffing, wellbeing, and operational costs using verified sources.

2.2 Organizational context

Describe the client context without claiming facts not supplied by the organization.

2.3 Business problem and stakeholders

Cover employees, HR, supervisors, operations, finance, senior management, and data governance stakeholders.

2.4 Consultancy objectives

Link objectives to RQ1-RQ10.

3. Research Questions and Hypotheses

Use alpha = 0.05. Final hypotheses must be selected after inspecting group sizes and assumptions. Examples include gender mean comparison, department mean differences, and numeric associations. For every test report H0, H1, method, assumptions, statistic, p-value, confidence interval/effect size where appropriate, decision, and business interpretation.

4. Literature Review

The supplied Research Papers/ folder contains 18 PDF files covering absenteeism determinants, occupational health, manufacturing, predictive modelling, machine learning, decision support, and workplace interventions. One file is

supplementary content for a JAMA Psychiatry article, so it should not be counted as a separate paper. The detailed inventory and comparison matrix are in literaturereview.md, and DOI/status verification is in citationaudit.md.

The literature indicates that absenteeism is associated with multiple worker, workplace, health, psychosocial, and non-work factors. Recent studies increasingly use machine learning, penalised regression, network analysis, historical attendance profiles, and decision-support tools. However, definitions of absence vary across studies, prediction does not establish causality, and individual-level HR prediction creates privacy, fairness, and punitive-misuse risks.

This project addresses the gap by modelling the supplied continuous AbsentHours outcome, auditing implausible records, comparing interpretable and regularised regression, and translating results into an ethical decision-support framework. Before submission, resolve the one unresolved publisher record and confirm the required peer-review and publication-date counts from the full papers.

5. Dataset and Methodology

Document the 13 observed variables, raw-file preservation, quality checks, excluded identifiers, sensitivity analysis for flagged records, 80/20 split, cross-validation, and leakage-safe preprocessing.

6. Descriptive Analysis

Insert outputs/tables/descriptivestatistics.csv, categoricalsummary.csv, and groupsummary.csv, plus figures. Explain operational implications without causal language.

7. Statistical Inference

Insert generated hypothesisresults.csv, posthocresults.csv, and correlationresults.csv. Prefer Welch/Kruskal-Wallis where assumptions or group imbalance require it. Report practical effect sizes, not p-values alone.

8. Predictive Statistical Modelling

Compare multiple regression, Ridge, LASSO, and Elastic Net. Report cross-validation MAE/RMSE, test MAE/RMSE/R2, coefficients, uncertainty for the inferential model, residual diagnostics, heteroscedasticity, independence, normality, and VIF. Do not select solely on RMSE.

8.1 Regression assumption diagnostics

The OLS diagnostic output is in outputs/tables/regressiondiagnostics.csv, with residual observations in outputs/tables/residualobservations.csv and figures in outputs/figures/regressionresidualdiagnostics.png and outputs/figures/regressionresidualqq.png.

For the numeric OLS baseline using Age and LengthService, R^2 is 0.6971 and adjusted R^2 is 0.6970. The Breusch-Pagan test is significant (LM = 266.90, $p < 0.001$), providing evidence of non-constant residual variance. The Jarque-Bera test is significant (JB = 116.69, $p < 0.001$), so residual normality is not supported. The Durbin-Watson statistic is 1.991; because the data are cross-sectional rather than ordered time-series observations, this is only a limited independence diagnostic and should not be interpreted as time-series evidence. The maximum Cook's distance is 0.0804 and 427 observations exceed the high-leverage screening threshold used in the diagnostic output. These observations should be investigated, not automatically deleted.

Because heteroscedasticity and non-normal residuals are present, conventional OLS p-values should be interpreted cautiously. The regularised models should be compared using cross-validation and held-out error, while any inferential coefficient claims should report limitations and, where required, heteroscedasticity-robust standard errors in the final

statistical appendix.

8.2 Age-record sensitivity analysis

The sensitivity outputs are outputs/tables/agesensitivitysummary.csv and outputs/tables/agesensitivitymodels.csv. Removing the 34 records flagged for implausible age or service-age inconsistency changes mean AbsentHours from 61.2840 to 61.5350, Pearson correlation between Age and AbsentHours from 0.8302 to 0.8331, and the numeric OLS R^2 from 0.6971 to 0.7020. The age coefficient changes from 4.1192 to 4.1992, while the statistical significance conclusion is unchanged.

For the full dataset, LASSO has the lowest test RMSE (25.956). After excluding the flagged records, Elastic Net has the lowest test RMSE (25.219), followed closely by LASSO (25.252). This indicates that the broad predictive conclusion is reasonably stable, but the exact regularised-model ranking is sensitive to the flagged records. The final recommendation should therefore report both analyses and avoid presenting one model as universally superior.

9. PCA Critical Evaluation

The current numerical feature set is small. Use pcaresults.csv and the cumulative-variance figure. Explain whether dimensionality reduction improves prediction enough to justify reduced interpretability.

10. Bayesian Methods

Naive Bayes is a classification method and is not directly suitable for continuous AbsentHours. Bayesian regression may add credible intervals, priors, hierarchical structure, and updating, but requires prior-sensitivity analysis. The implementation status is recorded in bayesianstatus.json; do not claim it was run unless it was.

11. Experimental Design

No experiment was conducted. Evaluate a future CRD comparing supportive intervention and control, and an RCBD blocking by department, division, job category, or location. Discuss contamination, ethics, feasibility, randomisation, and causal interpretation.

12. Time Series

No valid time variable exists in this dataset. Propose collection of dated department-level records and evaluate trend, seasonality, moving averages, ARIMA/SARIMA, forecast intervals, staffing, and early-warning use.

13. Innovation Proposal

The proposed innovation is the Employee Absenteeism Risk Intelligence and Decision Support Framework, described fully in conceptualframework.md. It converts validated employee and workforce records into aggregate absenteeism insights, model predictions, uncertainty information, and supportive management actions.

The business need is to help HR and operations understand where absenteeism pressure is concentrated and how staffing, workload, scheduling, and wellbeing resources could be allocated. The solution includes a data-quality and privacy layer, statistical insight layer, prediction layer, risk segmentation, role-based dashboard, supportive intervention workflow, and monitoring feedback loop.

Expected benefits include earlier identification of department-level staffing pressure, more targeted workforce planning, consistent reporting of uncertainty, better coordination between HR and operations, and a repeatable process for updating

the model when dated data become available. The solution is deliberately designed for organizational decision support rather than automatic employee surveillance or punishment.

Implementation challenges include invalid records, small group sizes, model drift, algorithmic bias, privacy, causal overinterpretation, and user adoption. Required resources include an HR owner, operations representative, statistician, data engineer, dashboard developer, privacy or ethics reviewer, secure storage, role-based access, version control, audit logging, and a data dictionary. A phased roadmap covers governance, data foundation, analytical pilot, dashboard pilot, intervention evaluation, and ongoing monitoring.

Organizational impact includes a shift from periodic descriptive reporting to governed, evidence-based workforce planning. Success should be assessed using data-quality resolution, model stability, subgroup performance, dashboard adoption, staffing-response time, supportive-intervention outcomes, employee feedback, and privacy incidents. A reduction in recorded absence alone should not be treated as proof of success.

14. Industry Expert Validation

Industry expert validation must be completed using expertfeedbacktemplate.md. The consultation should involve at least one real HR, manufacturing, workforce-planning, occupational-health, or relevant industry expert. The team must present the problem, selected findings, proposed framework, recommendations, and implementation risks, then record professional feedback about relevance, practicality, feasibility, and organizational impact.

The final report should include the expert's role and consultation date, consultation method, anonymised feedback summary, evidence reference, key themes, and a table showing which recommendations changed because of the feedback. Acceptable evidence includes an interview summary, meeting minutes, email communication, completed feedback form, or permitted meeting screenshot. Do not fabricate feedback or imply that the proposal was validated before the consultation occurred.

At the current stage, expert evidence is pending. The recommendations must therefore distinguish statistical evidence and literature evidence from unavailable expert evidence.

15-19. Recommendations, risks, limitations, future research, conclusion

Use recommendations.md. Recommendations must cover strategic action, operational action, risk management, ethics, and future research. Each recommendation must separately identify statistical evidence, literature evidence, expert evidence, implementation owner/resources, success measures, and remaining uncertainty. Mark expert evidence as pending until the real consultation is completed.

20. References

Bae, Y.-H. (2021). Exploratory analysis of related factors with absenteeism and presenteeism on workers: Using the Fourth Korea Working Condition Survey. *International Journal of Environmental Research and Public Health*. https://doi.org/10.3390/ijerph182111214

Bowen, F., Gentle-Genitty, C., Siegler, J., & Jackson, M. (2022). Revealing underlying factors of absenteeism: A machine learning approach. *Frontiers in Psychology*. https://doi.org/10.3389/fpsyg.2022.958748

Cikes, V., Maskarin Ribaric, H., & Crnjar, K. (2018). The determinants and outcomes of absence behavior: A systematic literature review. *Social Sciences*. https://doi.org/10.3390/socsci7080120

Duchaine, C. S., Aube, K., Gilbert-Ouimet, M., Vezina, M., Ndjaboue, R., Massamba, V., Talbot, D., Lavigne-Robichaud, M., Trudel, X., Pena-Gralle, A.-P. B., Lesage, A., Moore, L., Milot, A., Laurin, D., & Brisson, C. (2020). Psychosocial stressors at work and the risk of sickness absence due to a diagnosed mental disorder: A systematic review and meta-analysis. *JAMA Psychiatry*. https://doi.org/10.1001/jamapsychiatry.2020.0322

Kim, D., & Lee, J. W. (2025). Predicting absenteeism at workplace using machine learning and network analysis. *SAGE Open*. https://doi.org/10.1177/21582440251336019

Lawrance, N., Petrides, G., & Guerry, M.-A. (2021). Predicting employee absenteeism for cost effective interventions. *Decision Support Systems*. https://doi.org/10.1016/j.dss.2021.113539

Llamas Blazquez, P. (2025). Predicting workplace absenteeism using machine learning: A pilot study in occupational health. *Journal of Occupational Medicine and Toxicology*. https://doi.org/10.1186/s12995-025-00482-5

Nath, G., Wang, Y., Coursey, A., Saha, K. K., Prabhu, S., & Sengupta, S. (2022). Incorporating a machine learning model into a web-based administrative decision support tool for predicting workplace absenteeism. *Information, 13*, 320. https://doi.org/10.3390/info13070320

Nawata, K. (2024). Evaluation of physical and mental health conditions related to employees' absenteeism. *Frontiers in Public Health*. https://doi.org/10.3389/fpubh.2023.1326334

Nowak, J., Emmermacher, A., Wendsche, J., Dobler, A.-S., & Wegge, J. (2023). Presenteeism and absenteeism in the manufacturing sector: A multilevel approach identifying underlying factors and relations to health. *Current Psychology*. https://doi.org/10.1007/s12144-022-03013-1

Odeen, M., Magnussen, L. H., Maeland, S., Larun, L., Eriksen, H. R., & Tveito, T. H. (2013). Systematic review of active workplace interventions to reduce sickness absence. *Occupational Medicine, 63*, 7-16. https://doi.org/10.1093/occmed/kqs198

Saruan, N. A. M., Yusoff, H. M., Fauzi, M. F. M., Wan Puteh, S. E., et al. (2020). Unplanned absenteeism: The role of workplace and non-workplace stressors. *International Journal of Environmental Research and Public Health*. https://doi.org/10.3390/ijerph17176132

Thorsen, S. V., Flyvholm, M.-A., Pedersen, J., Bultmann, U., Andersen, L. L., & Bjorner, J. B. (2021). Associations between physical and psychosocial work environment factors and sickness absence incidence depend on the lengths of the sickness absence episodes: A prospective study of 27,678 Danish employees. *Occupational and Environmental Medicine, 78*, 46-53. https://doi.org/10.1136/oemed-2020-106554

Wagner, S., White, M., Schultz, I., Murray, E., Bradley, S. M., Hsu, V., McGuire, L., & Schulz, W. (2014). Modifiable worker risk factors contributing to workplace absence: A stakeholder-centred best-evidence synthesis of systematic reviews. *Work, 49*, 541-558. https://doi.org/10.3233/WOR-131709

White, M. I., Wagner, S. L., Schultz, I. Z., Murray, E., Bradley, S. M., Hsu, V., McGuire, L., & Schulz, W. (2015). Non-modifiable worker and workplace risk factors contributing to workplace absence: A stakeholder-centred synthesis of systematic reviews. *Work, 52*, 353-373. https://doi.org/10.3233/WOR-152134

Zupancic, P., & Panov, P. (2024). Predicting employee absence from historical absence profiles with machine learning. *Applied Sciences, 14*, 7037. https://doi.org/10.3390/app14167037

Alars, M., & Albakry, A. (2026). Data-driven predictive modelling of employee absenteeism using workflow automation platforms. *International Journal of Innovative Technology and Interdisciplinary Sciences, 9*(1), 115-136. https://doi.org/10.15157/IJITIS.2026.9.1.115-136

The Anzoom et al. paper in the supplied folder is excluded from this final list because its publisher, DOI, journal, and peer-review status remain unresolved. See citationaudit.md.

21. Appendices

Include data dictionary, validation output, model diagnostics, sensitivity analysis, expert evidence, contribution statement, and reproducibility instructions.

Literature Review: Employee Absenteeism

Evidence note

This review was prepared from the 18 PDF files supplied in Research Papers/. The papers address absenteeism determinants, occupational health, manufacturing, prediction, decision support, and workplace interventions. DOI and bibliographic verification is recorded in citationaudit.md. The review does not treat file names as citations, and it does not invent DOI, sample-size, or method details that were not visible in the supplied article text.

Introduction

Employee absenteeism is an organizational problem because absence can affect staffing continuity, workload distribution, productivity, service delivery, and employee wellbeing. However, absenteeism is not a single-factor outcome. The literature links absence to worker characteristics, physical and psychosocial working conditions, health, non-work stressors, job context, and organizational arrangements.

This creates two analytical challenges. Management needs to understand which observable groups or characteristics are associated with higher absence, while also ensuring that predictions are not treated as proof of employee fault. The literature therefore supports a consultancy approach that combines statistical inference, predictive modelling, organizational interpretation, and ethical governance. The present project follows this approach by analysing continuous AbsentHours, comparing interpretable and regularised regression, and evaluating how results could support workforce planning without punitive individual decisions.

Literature landscape

The supplied literature falls into four connected streams:

1. Determinants and risk factors: systematic reviews and occupational-health studies examine worker, workplace, physical, psychosocial, health, and non-work factors associated with absence.

2. Prediction and analytics: recent studies apply machine learning, penalised regression, association analysis, network analysis, and decision-support tools to absenteeism prediction.

3. Intervention and organizational action: intervention reviews examine ways of reducing sickness absence and emphasise that prediction should connect to feasible, ethical support.

4. Manufacturing and sector context: manufacturing-focused work uses multilevel or workplace-level approaches to account for clustered employees and health-related factors.

Paper inventory and relevance matrix

File	Verified title	Authors shown in supplied PDF	Journal/source shown	Year shown	Main relevance	Method/finding to verify in full text
1-s2.0-S2590005626000895-main.pdf	Predicting employee absenteeism: A framework and its validation	Rashid Anzoom; Syed Mithun Ali; Prattusha Bhattacharjee; Golam Kabir; Sanjoy Kumar Paul	Article PDF; journal metadata to verify	2026	Recent predictive framework and validation; highly relevant to the consultancy framework	Confirm dataset, target definition, algorithms, validation design, and final performance claims
IJITIS164.pdf	Data-Driven Predictive Modelling of Employee Absenteeism Using Workflow Automation Platforms	Mohammed Alars; Abbas Albakry	International Journal of Innovative Technology and Interdisciplinary Sciences	2026 issue; received/accepted dates shown	Directly relevant predictive modelling and workflow implementation; the abstract states an enterprise dataset of 8,336 employees	Verify peer-review status, dataset provenance, binary target definition, variables, and performance results
PredictingAbsenteeismatWorkplaceUsingMachine.pdf	Predicting Absenteeism at Workplace Using Machine Learning and Network Analysis	Donggeun Kim; Jai Woo Lee	SAGE Open	2025	Directly relevant; combines association analysis, odds ratios, network analysis, PCA, penalised regression, and imbalanced classification	Verify exact sample, variables, algorithms, validation, and whether PCA improved prediction
s12995-025-00482-5.pdf	Title and metadata to be transcribed from article cover	Authors and journal metadata to be transcribed	Journal article PDF	2025 shownin filename/cover context	Recent occupational-health evidence likely relevant to absenteeism and wellbeing	Read full article and verify title, objective, sample, method, and findings before citation

File	Verified title	Authors shown in supplied PDF	Journal/source shown	Year shown	Main relevance	Method/finding to verify in full text
PredictingEmployeeAbsencefromHistoricalAbsenc.pdf	Predicting Employee Absence from Historical Absence Profiles with Machine Learning	Peter Zupancic; Pan Panov	Applied Sciences	2024	Uses historical time-and-attendance and employee demographic data; relevant to future longitudinal/time-series data collection	Verify target construction, temporal split, models, and limitations
s12144-022-03013-1.pdf	Presenteeism and absenteeism in the manufacturing sector: A multilevel approach identifying underlying factors and relations to health	Joshua Nowak and co-authors	Current Psychology	2023	Direct manufacturing-sector relevance; supports multilevel thinking and health-related interpretation	Verify sample, hierarchical structure, variables, model specification, and findings
information-13-00320.pdf	Incorporating a Machine Learning Model into a Web-Based Administrative Decision Support Tool for Predicting Workplace Absenteeism	Gopal Nath; Yawei Wang; Austin Coursey; Krishna K. Saha; Srikanth Prabhu; Saptarshi Sengupta	Information	2022	Supports the proposed dashboard and decision-support framework	Verify data, target definition, model comparison, interface, validation, and practical limitations

File	Verified title	Authors shown in supplied PDF	Journal/source shown	Year shown	Main relevance	Method/finding to verify in full text
oemed-2020-106554.pdf	Associations between physical and psychosocial work environment factors and sickness absence incidence depend on the lengths of the sickness absence episodes: a prospective study of 27,678 Danish employees	Sannie V. Thorsen; Mari-Ann Flyvholm; Jacob Pedersen; Ute Bultmann; Lars L. Andersen; Jakob Bue Bjorner	Occupational and Environmental Medicine	2021	Strong evidence that work-environment associations may vary by absence-episode length; supports cautious interpretation of a single continuous target	Verify statistical model, episode definitions, covariates, and limitations
Predictingemployeeabsenteeismforcosteffective.pdf	Predicting employee absenteeism for cost effective interventions	Natalie Lawrance; George Petrides; Marie-Anne Guerry	Decision Support Systems	2021	Directly links prediction to cost-effective intervention rather than accuracy alone	Verify dataset, cost model, algorithms, validation, and intervention recommendations
fpubh-11-1326334.pdf	Evaluation of physical and mental health conditions related to employees absenteeism	Kazumitsu Nawata	Frontiers in Public Health	2023 issue indicated by filename	Supports health and wellbeing interpretation of absence-related variables	Verify study design, population, health variables, statistical methods, and causal limitations
ijerph-18-11214.pdf	Exploratory Analysis of Related Factors with Absenteeism and Presenteeism on Workers: Using the Fourth Korea Working Condition Survey	Young-Hyeon Bae and co-authors	International Journal of Environmental Research and Public Health	2021 issue indicated by volume	Supports exploratory factor analysis using worker and workplace conditions	Verify sample, survey design, regression/association methods, and findings

File	Verified title	Authors shown in supplied PDF	Journal/source shown	Year shown	Main relevance	Method/finding to verify in full text
ijerph-17-06132.pdf	Unplanned Absenteeism: The Role of Workplace and Non-Workplace Stressors	Nur Adibah Mat Saruan; Hanizah Mohd Yusoff; Mohd Fadhli Mohd Fauzi; Sharifa Ezat Wan Puteh and co-authors	International Journal of Environmental Research and Public Health	2020 issue indicated by volume	Supports inclusion of workplace and non-workplace stressors in the research landscape	Verify sample, measures, model, and study limitations
fpsyg-13-958748.pdf	Revealing underlying factors of absenteeism: A machine learning approach	Francis Bowen; Carolyn Gentle-Genitty; Janaina Siegler; Marlin Jackson	Frontiers in Psychology	2022 issue indicated by article identifier	Supports machine-learning exploration of absenteeism factors	Verify dataset, feature construction, algorithms, validation, and interpretability
socsci-07-00120.pdf	The Determinants and Outcomes of Absence Behavior: A Systematic Literature Review	Vedrana Cikes; Helga Maskarin Ribaric; Kristina Crnjar	Social Sciences	2018 issue indicated by volume	Provides broad theoretical synthesis of determinants and outcomes	Verify review search strategy, included studies, findings, and research gaps
wagner-et-al-2014-modifiable-worker-risk-factors-contributing-to-workplace-absence-a-stakeholder-centred-best-evidence.pdf	Modifiable worker risk factors contributing to workplace absence: A stakeholder-centred best-evidence synthesis of systematic reviews	Shannon Wagner; Marc White; Izabela Z. and co-authors	Work	2014	Supports stakeholder-centred identification of modifiable risk factors and intervention design	Verify complete author list, review method, evidence grading, and findings

File	Verified title	Authors shown in supplied PDF	Journal/source shown	Year shown	Main relevance	Method/finding to verify in full text
white-et-al-2015-non-modifiable-worker-and-workplace-risk-factors-contributing-to-workplace-absence-a-stakeholder.pdf	Non-modifiable worker and workplace risk factors contributing to workplace absence: A stakeholder-centred synthesis of systematic reviews	Marc I. White; Shannon L. Wagner; Izabela Z. and co-authors	Work	2015	Supports separation of non-modifiable characteristics from intervention targets	Verify complete author list, review method, risk-factor categories, and findings
kqs198.pdf	Systematic review of active workplace interventions to reduce sickness absence	M. Odeen; L. H. Magnussen; S. Maeland; L. Larun; H. R. Eriksen; T. H. Tveito	Occupational Medicine	2013	Supports CRD/RCBD discussion and the need to connect analytics to evaluated interventions	Verify included interventions, evidence quality, outcomes, and conclusions
yoi200013supp1prod.pdf	Supplementary content for *Psychosocial Stressors at Work and the Risk of Sickness Absence Due to a Diagnosed Mental Disorder: A Systematic Review and Meta-analysis*	C. S. Duchaine; K. Aube; M. Gilbert-Ouimet and co-authors	JAMA Psychiatry supplementary content	2020	Supports psychosocial-risk and mental-health context	Cite the main article once using DOI 10.1001/jamapsychiatry.2020.0322; do not count the supplement as a separate paper

Critical synthesis

Methodological patterns

The papers show a movement from descriptive and occupational-health research toward predictive analytics and decision-support systems. Recent prediction studies use machine learning, penalised regression, association analysis, network analysis, PCA, historical absence profiles, and web-based tools. This supports comparing interpretable linear regression with Ridge, LASSO, and Elastic Net in the present consultancy rather than treating accuracy as the only

criterion.

Determinants and risk factors

The review literature presents absenteeism as a multidimensional organizational outcome. Wagner et al. focus on modifiable worker risk factors, while White et al. distinguish non-modifiable worker and workplace factors. Cikes et al. provide a broader synthesis of determinants and outcomes. Together, these papers imply that organizations should distinguish between characteristics that describe differences in absence and factors that can realistically be changed through intervention.

The occupational-health studies extend this argument by emphasizing physical, psychosocial, and health-related conditions. Thorsen et al. report that work-environment associations may depend on the length of absence episodes. Saruan et al. examine workplace and non-workplace stressors, while Nawata considers physical and mental health conditions. These findings suggest that a single aggregate measure such as AbsentHours may combine different types of absence. The present analysis should therefore treat AbsentHours as an overall outcome and identify episode type, cause, and time period as important future data requirements.

Prediction and decision support

The prediction studies show increasing interest in machine learning and operational decision support. Bowen et al. use machine learning to investigate underlying absenteeism factors. Kim and Lee combine association analysis, network analysis, PCA, and penalised modelling. Zupancic and Panov focus on historical absence profiles, which is relevant to future longitudinal attendance data. Nath et al. connect prediction to a web-based administrative decision-support tool, while Lawrance et al. frame prediction around cost-effective interventions.

These studies justify comparing several models in the present project. Regularised models may improve predictive stability when categorical variables are encoded into many related predictors, while multiple linear regression provides a more direct inferential baseline. The final recommendation should therefore consider test error, cross-validation stability, coefficient behaviour, assumptions, transparency, and the consequences of operational use.

Manufacturing and multilevel context

Nowak et al. address absenteeism and presenteeism in manufacturing using a multilevel perspective. This is relevant because employees may be clustered within departments, locations, job categories, or work environments. Employees in the same unit may share schedules, supervisors, workloads, and physical conditions.

The current dataset does not provide enough repeated or contextual information to claim a full multilevel causal analysis. It does contain department, division, store location, business unit, and job-title variables. The project can therefore conduct cautious group comparisons and regression modelling while identifying hierarchical modelling as a future extension.

Interventions and experimental evidence

Odeen et al. review active workplace interventions to reduce sickness absence. This is important because prediction is useful only when it informs a feasible response. A future CRD could evaluate a supportive intervention against a control condition, while an RCBD could randomise within departments or locations to account for known organizational differences.

The intervention literature also creates an ethical constraint. Absence-reduction interventions should not become surveillance or punishment. They should address staffing, workload, wellbeing, work design, or access to support, with safeguards against coercion, contamination, unequal access, and stigmatization.

Methodological tensions

Four tensions recur across the literature:

1. Explanation versus prediction: occupational-health studies often prioritize interpretable associations, while machine-learning studies may prioritize prediction performance.

2. Continuous versus categorical outcomes: binary high-risk labels simplify decisions but can discard information and introduce threshold sensitivity.

3. Individual versus organizational intervention: individual predictions may be operationally attractive but create greater privacy and fairness risks than department-level planning.

4. Cross-sectional versus longitudinal evidence: cross-sectional data identify associations, while historical and prospective data are needed for temporal ordering, recurrence, forecasting, and intervention evaluation.

The present consultancy retains these tensions rather than claiming that one method is universally superior. It uses a continuous target, compares regularised regression with an interpretable baseline, evaluates PCA rather than assuming it is necessary, and treats Bayesian, time-series, and experimental methods as conditional extensions.

Key strengths in prior work

- Large or longitudinal samples in several occupational-health studies.

- Use of prospective or multilevel designs in some studies.

- Explicit connection between prediction and intervention in decision-support research.

- Recognition that absence is influenced by health, psychosocial, physical, workplace, and non-work factors.

Key limitations and gaps

- Many studies use different definitions of absence, making direct comparison difficult.

- Classification targets may discard information contained in a continuous absence-hours measure.

- Predictive studies may not provide enough explanation for HR decision-making.

- Cross-sectional employee data cannot establish causality.

- Individual-level prediction raises privacy, fairness, and punitive-misuse risks.

- The current dataset has no time variable, so temporal forecasting cannot be claimed.

- The current dataset contains implausible ages that require sensitivity analysis.

Contribution of this consultancy project

This project addresses the identified gap by combining: (1) explicit data-quality auditing, (2) continuous AbsentHours modelling, (3) robust group comparisons, (4) interpretable and regularised regression, (5) critical PCA/Bayesian evaluation, and (6) an ethical decision-support proposal. It should report associations and predictive performance without claiming that demographic or organizational variables cause absence.

Research gap and project contribution

The reviewed studies leave a practical gap for an organization that has employee-level records but lacks detailed absence dates, absence causes, intervention history, and repeated measurements. Such an organization needs an approach that is statistically defensible but understandable to HR and operations managers. It must also separate descriptive evidence from causal claims and prevent individual-level predictions from becoming automatic disciplinary decisions.

This consultancy addresses that gap by combining data-quality auditing, continuous AbsentHours modelling, robust group comparisons, interpretable and regularised regression, critical evaluation of PCA and Bayesian methods, and an ethical decision-support proposal. The contribution is not to claim a universal absenteeism model. It is to demonstrate a reproducible workflow for deciding what the available data can support, what it cannot support, and what additional data or

experimental design would be required for stronger organizational decisions.

Analytical framework justified by the literature

The literature supports the following sequence:

1. Audit the dataset and identify invalid or implausible records.
2. Describe the distribution of AbsentHours and the composition of employee groups.
3. Test differences between meaningful groups using robust methods when variances or group sizes are unequal.
4. Examine age and service-length relationships using Pearson or Spearman correlation as appropriate.
5. Compare multiple linear regression with Ridge, LASSO, and Elastic Net using leakage-safe preprocessing and cross-validation.
6. Evaluate residual assumptions, multicollinearity, model stability, and sensitivity to flagged records.
7. Translate reliable results into aggregate workforce planning and supportive intervention recommendations.

This framework is consistent with the literature's combined emphasis on risk-factor understanding, predictive validation, intervention feasibility, and ethical organizational use.

Citation audit before submission

- Use citationaudit.md as the record of DOI and metadata verification.
- Confirm journal peer-review policies or article types before counting sources toward the assignment requirement.
- Resolve the Anzoom et al. publisher record before citing or counting that paper.
- Confirm that at least five papers fall within the lecturer's definition of the last five years.
- Read each paper's method and results sections before stating numerical findings.
- Add the final reference style consistently to the consultancy report.

Employee Absenteeism Risk Intelligence and Decision Support Framework

Proposed solution

The proposed innovation is the Employee Absenteeism Risk Intelligence and Decision Support Framework. It is a privacy-aware analytics service that converts employee and workforce records into aggregate absenteeism insights, validated predictions, and supportive management actions.

The framework is designed for HR and operations management rather than automatic employee surveillance. Its primary decision unit should be the department, division, business unit, or location. Individual-level scores, if ever considered, require separate governance approval, strong justification, access restriction, explanation, and human review.

Business need

The organization needs to understand where absenteeism pressure is concentrated, which observable characteristics are associated with AbsentHours, and how staffing or wellbeing resources could be allocated. A static report is insufficient because managers need recurring monitoring, uncertainty information, model-performance checks, and a feedback mechanism.

The current analysis shows that the available data can support descriptive group comparisons and predictive modelling, but it cannot establish causality or genuine time trends. The solution therefore combines current evidence with a data-collection roadmap for dated absenteeism records.

```
flowchart TD A[Employee and workforce data] --> B[Data quality and privacy layer] B --> C[Exploratory statistical analysis] C --> D[Statistical inference] C --> E[Predictive regression models] D --> F[Evidence and uncertainty register] E --> G[Model validation and fairness checks] F --> H[Risk intelligence dashboard] G --> H H --> I[Supportive workforce intervention] I --> J[Outcome monitoring] J --> K[New dated data] K --> B J -. expert and employee feedback .-> H
```

Solution components

1. Data layer: collect only necessary employee and organizational variables; preserve the raw source and record data lineage.

2. Quality and privacy layer: check impossible ages, service inconsistencies, duplicates, missingness, outliers, access permissions, and retention rules.

3. Statistical insight layer: produce distributions, group summaries, robust tests, correlations, effect sizes, and uncertainty intervals.

4. Prediction layer: compare interpretable regression with regularised models and report cross-validation and held-out performance.

5. Risk segmentation layer: classify aggregate operational pressure into monitoring bands such as stable, watch, and priority, using validated thresholds rather than arbitrary individual labels.

6. Dashboard layer: show department-level absenteeism, group comparisons, model drivers, uncertainty, data-quality warnings, and recommended supportive actions.

7. Intervention layer: connect findings to staffing review, workload assessment, wellbeing support, scheduling review, and resource allocation.

8. Monitoring layer: track outcomes, model drift, subgroup performance, fairness, false positives, missingness, and user feedback.

Dashboard design

The management dashboard should contain:

- Executive view: organization-wide mean, median, zero-absence proportion, data-quality status, and current model performance.
- Department view: department counts, mean and median AbsentHours, uncertainty intervals, sample sizes, and comparison warnings for small groups.
- Workforce planning view: staffing pressure, locations or units requiring operational review, and scenario notes.
- Driver view: validated model coefficients or importance proxies with plain-language interpretation and a clear association-not-causation warning.
- Governance view: missingness, flagged records, fairness checks, drift indicators, access logs, model version, and last review date.

The dashboard should avoid displaying names, surnames, or raw individual predictions by default. It should support decision-making at an organizational level and provide links to data-quality and model documentation.

Expected benefits

Strategic benefits

- Evidence-based workforce planning instead of uniform interventions.
- Earlier identification of department-level staffing pressure.
- Better alignment between HR, operations, wellbeing, and resource decisions.
- A repeatable analytical process that can be updated when dated records become available.

Operational benefits

- Faster identification of unusual group patterns.
- More targeted review of workload, scheduling, staffing, and working conditions.
- Consistent reporting of uncertainty and data-quality limitations.
- Reduced risk of selecting a model solely because it has the lowest prediction error.

Employee and organizational benefits

- Greater emphasis on supportive interventions rather than punishment.
- Transparent model limitations and human oversight.
- Opportunity to correct inaccurate employee records.
- Better evidence for evaluating future absenteeism-reduction interventions.

Implementation roadmap

Phase	Activities	Deliverable
1. Governance and discovery	Confirm business owner, permitted uses, data access, retention, and employee safeguards	Approved use and governance register
2. Data foundation	Add dated absence records, define measures, validate existing fields, and document lineage	Quality-controlled data model
3. Analytical pilot	Re-run descriptive analysis, inference, models, sensitivity checks, and subgroup fairness checks	Validated pilot report
4. Dashboard pilot	Build role-based aggregate dashboard with uncertainty and warnings	HR/operations pilot dashboard
5. Intervention evaluation	Select supportive actions and evaluate them using CRD or RCBD where ethical and feasible	Intervention evaluation report
6. Monitoring and update	Review drift, outcomes, fairness, data quality, and stakeholder feedback	Model monitoring and update cycle

Required resources

- People: HR subject-matter owner, operations representative, data analyst/statistician, data engineer, dashboard developer, privacy or ethics reviewer, and employee representative where appropriate.

- Data: validated employee master data, dated absence records, department/location structure, staffing counts, and intervention records where lawful and necessary.

- Technology: secure data storage, Python or R analytical environment, version control, dashboard platform, role-based access control, and audit logging.

- Process: data dictionary, model card, approval workflow, incident response process, review calendar, and employee correction mechanism.

- Time: staged pilot delivery followed by monitored expansion; exact cost requires organizational estimates and should not be fabricated in the report.

Implementation challenges and controls

Challenge	Risk	Control
Implausible or incomplete records	Biased results and wrong interventions	Automated validation, manual review, sensitivity analysis, and correction workflow
Small organizational groups	Unstable estimates and misleading rankings	Minimum sample warnings, aggregation, confidence intervals, and suppression rules
Model drift	Performance deterioration as workforce conditions change	Scheduled validation and trigger-based retraining
Algorithmic bias	Unequal treatment or discriminatory decisions	Subgroup performance checks, human review, and prohibited-use policy
Privacy and surveillance concerns	Loss of trust and inappropriate employee monitoring	Data minimization, role-based access, aggregation, transparency, and retention limits
Causal overinterpretation	Punitive or ineffective action	Association language, uncertainty display, and intervention evaluation
Adoption resistance	Dashboard ignored or misused	Co-design with HR and operations, training, feedback, and clear accountability

Organizational impact

The framework changes absenteeism management from periodic descriptive reporting to a governed decision-support process. HR gains a structured way to identify support needs, operations gains better evidence for staffing and workload planning, and senior management gains a repeatable view of risk and uncertainty. The organization also accepts new responsibilities: it must protect employee privacy, explain analytical outputs, monitor fairness, and ensure that predictions do not become automatic disciplinary decisions.

Success measures

The pilot should be evaluated using:

- Data-quality issue detection and resolution rate.
- Dashboard adoption by intended management users.
- Forecast or prediction error on future held-out records.
- Stability of model performance across departments and time periods.
- Change in staffing-planning response time.
- Number and outcome of supportive interventions.
- Employee and manager feedback.
- Privacy incidents, inappropriate-access events, and model-governance exceptions.

Success should not be defined only as a reduction in recorded absenteeism. A reduction could reflect improved attendance, changed recording practices, or harmful pressure on employees. Interpretation requires operational, employee, and governance evidence together.

Ethical operating rules

- Do not use predictions as sole evidence for discipline, dismissal, promotion, or denial of opportunity.
- Do not expose names or individual scores in the default management dashboard.
- Do not infer medical conditions or personal causes from the supplied variables.
- Allow employees and managers to challenge inaccurate data.
- Require human review for every consequential action.
- Reassess the framework when new variables, policies, or data sources are introduced.

Consultancy Recommendations

Evidence status

Statistical evidence is generated in outputs/. Literature and expert evidence must be added only after verification and real consultation.

Every final recommendation must have three evidence fields. A recommendation is not fully supported until all available fields are completed or the missing evidence is explicitly reported as a limitation.

Strategic

Use department-level absenteeism intelligence to support workforce planning and targeted investigation. Do not treat model output as proof of employee fault or cause.

Operational

Monitor department and business-unit summaries, staffing pressure, and absenteeism trends. Prioritise groups with statistically reliable differences and meaningful effect sizes.

HR

Use findings to guide supportive wellbeing, workload, scheduling, and return-to-work conversations. Apply human review and allow employees to challenge inaccurate records.

Data

Validate age and service fields, record collection dates in future systems, define absenteeism measures, and document data lineage and retention.

Technology

Implement a role-based dashboard showing trends, uncertainty, group summaries, model performance, and explanations. Avoid exposing unnecessary personally identifiable information.

Risk management

Monitor drift, calibration, subgroup performance, missingness, and false positives. Establish model review and rollback procedures.

Ethical

Use predictions for support and resource allocation, not automatic punishment, surveillance, or discrimination. Minimise personal data and secure access.

Future research

Conduct sensitivity analyses, collect longitudinal data, validate with HR experts, and evaluate supportive interventions using a pre-registered CRD or blocked RCBD where ethical and feasible.

Evidence mapping table

Recommendation	Statistical evidence	Literature evidence	Expert evidence	Status
Supportive department-level planning	groupsummary.csv, hypothesisresults.csv, modelcomparison.csv	Literature review: decision support, manufacturing, and intervention evidence	Complete after expert consultation	Statistical and literature evidence available; expert evidence pending
Improve data quality	datavalidation.json, agesensitivitysummary.csv, regressiondiagnostics.csv	Literature review: measurement and risk-factor limitations	Complete after expert consultation	Statistical evidence available; expert evidence pending
Use regularised prediction with human oversight	modelcomparison.csv, agesensitivitymodels.csv	Literature review: predictive modelling and ethical limitations	Complete after expert consultation	Statistical and literature evidence available; expert evidence pending
Collect dated department-level records	No valid time variable in current data	Literature review: historical profiles and longitudinal occupational studies	Complete after expert consultation	Future proposal; expert evidence pending
Avoid punitive individual predictions	Observational design and model limitations	Literature review: risk factors, interventions, and ethical decision support	Complete after expert consultation	Ethical requirement; expert evidence pending

Recommendation writing format

For each final recommendation, use this structure:

Recommendation: State the action for senior management.

Statistical evidence: Cite the generated table, figure, test, effect size, model result, or limitation.

Literature evidence: Cite the verified paper(s) and explain the relevant finding or methodological lesson.

Industry expert evidence: Summarise the real expert's feedback and identify the evidence record.

Implementation: State owner, resources, timing, and success measure.

Risk and ethics: State privacy, fairness, misuse, uncertainty, and human-oversight controls.

Citation Audit

Audit date: 2026-09-14

Audit rule

DOI metadata was checked against Crossref where available, and article metadata was checked against the supplied PDF. A DOI registry confirms bibliographic identity and publication type; it does not by itself prove peer review. The status below therefore distinguishes final journal articles from supplementary content and records where the journal's peer-review status must be confirmed from its policy.

Verified records

File	Verified citation identity	DOI	Publication status	Peer-review status
1-s2.0-S2590005626000895-main.pdf	*Predicting employee absenteeism: A framework and its validation*; Anzoom,Ali, Bhattacharjee, Kabir, Paul; journal and final DOI not resolved from Crossref	Not resolved	Article PDF; do not cite until publisher metadata is confirmed	Unverified; exclude from final count until verified
IJITIS164.pdf	*Data-Driven Predictive Modelling of Employee Absenteeism Using Workflow Automation Platforms*; Alars and Albakry; International Journal of Innovative Technology and Interdisciplinary Sciences, 2026 issue	10.15157/IJITIS.2026.9.1.115-136	Final research article identified in PDF; DOI not found in Crossref API	Journal article; confirm the journal's peer-review policy before counting
PredictingAbsenteeismatWorkplaceUsingMachine.pdf	Kim and Lee, *Predicting Absenteeism at Workplace Using Machine Learning and Network Analysis*, SAGE Open, 2025	10.1177/21582440251336019	Final journal article; Crossref metadata verified	SAGE Open journal article; peer-review status should be confirmed from journal policy

File	Verified citation identity	DOI	Publication status	Peer-review status
s12995-025-00482-5.pdf	Llamas Blazquez, *Predicting workplace absenteeism using machine learning: a pilot study in occupational health*, Journal of Occupational Medicine and Toxicology, 2025	10.1186/s12995-025-00482-5	Final open-access journal article; Crossref metadata verified	Journal research article; peer-review status should be confirmed from journal policy
PredictingEmployeeAbsencefromHistoricalAbsenc.pdf	Zupancic and Panov, *Predicting Employee Absence from Historical Absence Profiles with Machine Learning*, Applied Sciences, 2024	10.3390/app14167037	Final journal article; Crossref metadata verified	Journal article; peer-review status should be confirmed from journal policy
s12144-022-03013-1.pdf	Nowak, Emmermacher, Wendsche, Dobler, and Wegge, *Presenteeism and absenteeism in the manufacturing sector: A multilevel approach identifying underlying factors and relations to health*, Current Psychology, 2023	10.1007/s12144-022-03013-1	Final journal article; Crossref metadata verified	Springer journal article; peer-review status should be confirmed from journal policy

File	Verified citation identity	DOI	Publication status	Peer-review status
information-13-00320.pdf	Nath, Wang, Coursey, Saha, Prabhu, and Sengupta, *Incorporating a Machine Learning Model into a Web-Based Administrative Decision Support Tool for Predicting Workplace Absenteeism*, Information, 2022	10.3390/info13070320	Final journal article; Crossref metadata verified	Journal article; peer-review status should be confirmed from journal policy
oemed-2020-106554.pdf	Thorsen, Flyvholm, Pedersen, Bultmann, Andersen, and Bjorner, *Associations between physical and psychosocial work environment factors and sickness absence incidence depend on the lengths of the sickness absence episodes*, Occupational and Environmental Medicine, 2021	10.1136/oemed-2020-106554	Final journal article; Crossref metadata verified	BMJ journal research article; peer-review status should be confirmed from journal policy
Predictingemployeeabsenteeismforcosteffective.pdf	Lawrance, Petrides, and Guerry, *Predicting employee absenteeism for cost effective interventions*, Decision Support Systems, 2021	10.1016/j.dss.2021.113539	Supplied file is a journal pre-proof; final DOI record verified	Final journal article; peer-review status should be confirmed from journal policy

File	Verified citation identity	DOI	Publication status	Peer-review status
fpubh-11-1326334.pdf	Nawata, *Evaluation of physical and mental health conditions related to employees absenteeism*, Frontiers in Public Health, 2024 publication record	10.3389/fpubh.2023.1326334	Final journal article; Crossref metadata verified	Frontiers research article; peer-review status should be confirmed from journal policy
ijerph-18-11214.pdf	Bae, *Exploratory Analysis of Related Factors with Absenteeism and Presenteeism on Workers: Using the Fourth Korea Working Condition Survey*, International Journal of Environmental Research and Public Health, 2021	10.3390/ijerph182111214	Final journal article; Crossref metadata verified	Journal article; peer-review status should be confirmed from journal policy
ijerph-17-06132.pdf	Saruan, Yusoff, Fauzi, Wan Puteh, and co-authors, *Unplanned Absenteeism: The Role of Workplace and Non-Workplace Stressors*, International Journal of Environmental Research and Public Health, 2020	10.3390/ijerph17176132	Final journal article; DOI resolved through Crossref title search	Journal article; peer-review status should be confirmed from journal policy

File	Verified citation identity	DOI	Publication status	Peer-review status
fpsyg-13-958748.pdf	Bowen, Gentle-Genitty, Siegler, and Jackson, *Revealing underlying factors of absenteeism: A machine learning approach*, Frontiers in Psychology, 2022	10.3389/fpsyg.2022.958748	Final journal article; Crossref metadata verified	Frontiers research article; peer-review status should be confirmed from journal policy
socsci-07-00120.pdf	Cikes, Maskarin Ribaric, and Crnjar, *The Determinants and Outcomes of Absence Behavior: A Systematic Literature Review*, Social Sciences, 2018	10.3390/socsci7080120	Final journal article; DOI resolved through Crossref title search	Journal review article; peer-review status should be confirmed from journal policy
wagner-et-al-2014-modifiable-worker-risk-factors-contributing-to-workplace-absence-a-stakeholder-centred-best-evidence.pdf	Wagner, White, Schultz, Murray, Bradley, Hsu, McGuire, and Schulz, *Modifiable worker risk factors contributing to workplace absence: A stakeholder-centred best-evidence synthesis of systematic reviews*, Work, 2014	10.3233/WOR-131709	Final journal review article; Crossref metadata verified	Journal review article; peer-review status should be confirmed from journal policy

File	Verified citation identity	DOI	Publication status	Peer-review status
white-et-al-2015-non-modifiable-worker-and-workplace-risk-factors-contributing-to-workplace-absence-a-stakeholder.pdf	White, Wagner, Schultz, Murray, Bradley, Hsu, McGuire, and Schulz, *Non-modifiable worker and workplace risk factors contributing to workplace absence: A stakeholder-centred synthesis of systematic reviews*, Work, 2015	10.3233/WOR-152134	Final journal review article; Crossref metadata verified	Journal review article; peer-review status should be confirmed from journal policy
kqs198.pdf	Odeen, Magnussen, Maeland, Larun, Eriksen, and Tveito, *Systematic review of active workplace interventions to reduce sickness absence*, Occupational Medicine, 2013	10.1093/occmed/kqs198	Final journal review article; Crossref metadata verified	Journal review article; peer-review status should be confirmed from journal policy
yoi200013supp1prod.pdf	Supplementary online content for Duchaine et al., *Psychosocial Stressors at Work and the Risk of Sickness Absence Due to a Diagnosed Mental Disorder: A Systematic Review and Meta-analysis*, JAMA Psychiatry, 2020	10.1001/jamapsychiatry.2020.0322	Supplement, not a separate article; main article metadata verified through PubMed	Count the main JAMA Psychiatry article once, not the supplement as an additional paper

Final count interpretation

- Supplied PDF files: 18.

- The folder contains 17 regular article PDFs and one supplementary PDF. Replacing the supplement with its verified JAMA Psychiatry main-article citation gives 18 article-level sources without counting the supplement as a separate paper.

- One source remains unresolved for final citation: the Anzoom et al. PDF. Do not count it until its final publisher record, DOI, journal, and peer-review status are confirmed.

- At least 16 article-level sources are currently identified from journal records, subject to the lecturer's definition of peer-reviewed paper.

- Recent sources dated 2021 or later are present, but the final last five years count must follow the assignment's deadline rule.

Required final checks

1. Confirm journal peer-review policies or use the journal's published article type as evidence.

2. Replace any remaining author truncation with the complete author list required by the referencing style.

3. Confirm page range, volume, issue, and article number from each final journal record.

4. Add only verified DOI strings to the final reference list.

5. Do not cite the JAMA supplementary PDF as a separate research paper.

Appendix: Generated Statistical Outputs

Descriptive statistics

Variable	count	mean	median	std	var	min	max	skew	Q1	Q3	IQR
Age	8336.0	42.0071	42.1149	9.9398	98.7996	3.5047	77.938	-0.0199	35.2987	48.6669	13.3682
LengthService	8336.0	4.7829	4.6002	2.463	6.0663	0.0121	43.7352	5.7832	3.5759	5.6239	2.048
AbsentHours	8336.0	61.284	56.0058	49.0384	2404.7612	0.0	272.5301	0.6422	19.1276	94.2847	75.1571

Model comparison

Model	CVMAE	CVRMSE	TestMAE	TestRMSE	TestR2	Interpretability
LASSO	20.2149	25.5487	20.293	25.9558	0.7032	Moderate
ElasticNet	20.2508	25.5866	20.3	25.9664	0.7029	Moderate
Ridge	20.3577	25.7398	20.5183	26.2293	0.6969	Moderate
MultipleLinearRegression	20.7931	26.3642	21.1853	27.8867	0.6574	High

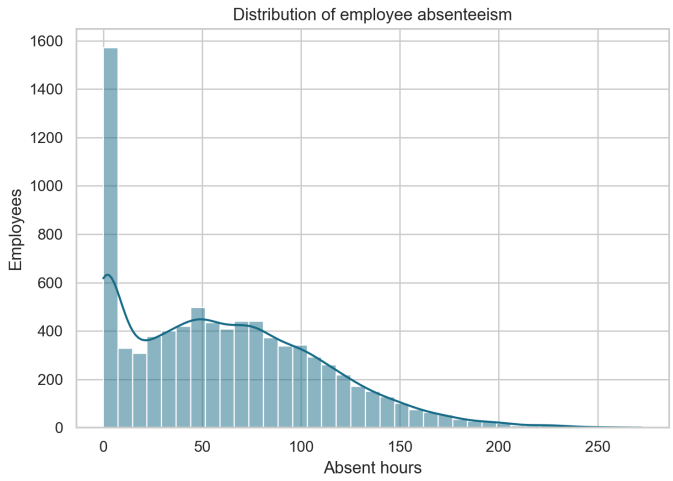
Age sensitivity analysis

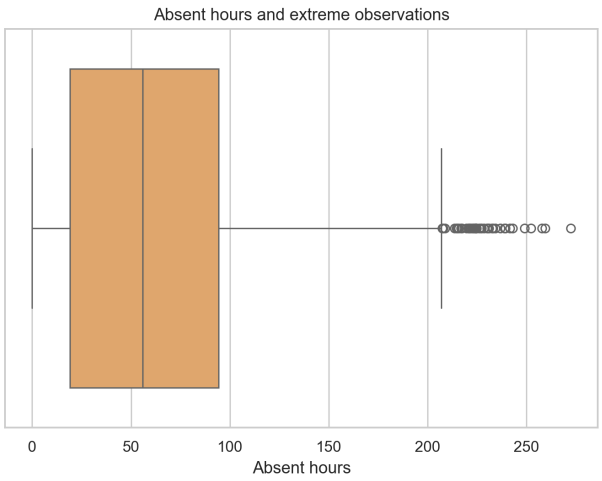
Analysis Set	Records	FlaggedRecordsExcluded	MeanAbsentHours	MedianAbsentHours	AgeAbsentHoursPearsonR	AgeCoefficient	AgePValue	OLS R2
All records	8336	0	61.284	56.0058	0.8302	4.1192	0.0	0.6971
Exclude flagged age/service records	8302	34	61.535	56.3026	0.8331	4.1992	0.0	0.702

Regression diagnostics

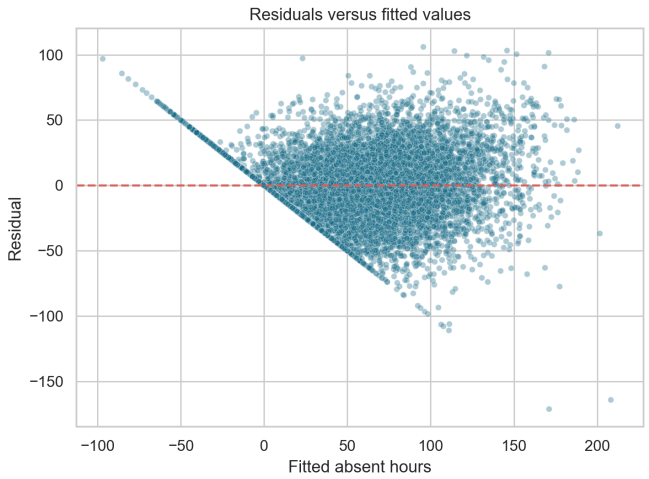
Model	Observations	R2	Adjusted R2	BreuschPaganLM	BreuschPaganPValue	BreuschPaganF	BreuschPaganFPValue	JarqueBera	JarqueBeraPValue	ResidualSkewness	ResidualKurtosis	DurbinWatson	MaxCookDistance	HighLeverageCount
OLS: Age + LengthService	8336	0.6971	0.697	266.8982	0.0	137.8135	0.0	116.6891	0.0	0.0928	3.5491	1.9912	0.0804	427

Absent-hours distribution



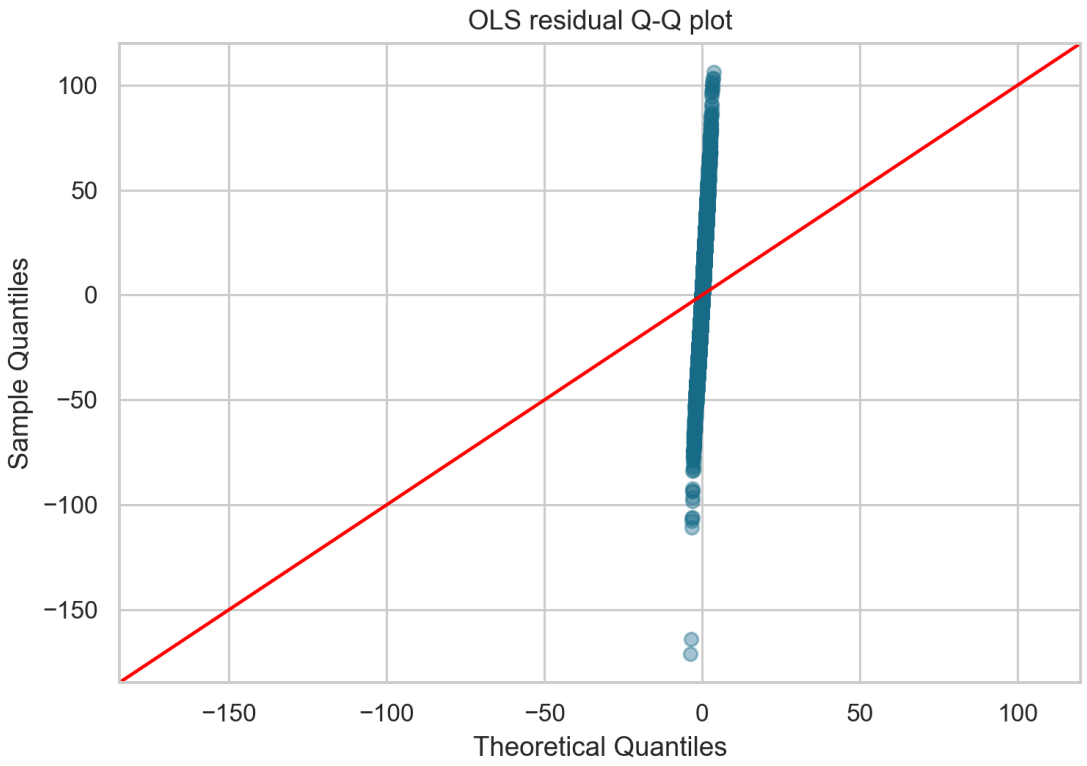


Regression residual diagnostics





Regression residual Q-Q plot


OLS residual Q-Q plot
Sample Quantiles
Theoretical Quantiles
100
50
0
−50
−100
−150
−150
−100
−50
0
50
100